CS4246/5446 · GROUP 23

Carbon-Aware Smart Building Energy Controller

Learning *when* to cool an office so the grid burns less carbon

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Reinforcement Learning and Sequential Decision Making · 12 Nov 2026

MOTIVATION

When a building uses power matters as much as how much

- Buildings use about 30% of global final energy; in Singapore, air-conditioning dominates commercial electricity
- Grid carbon intensity swings hour to hour with wind and solar output
- A building's thermal mass stores cooling, so load can be moved in time
- But an action now affects comfort hours later, which makes this a sequential decision problem

> Can an RL controller cut HVAC carbon beyond a well-tuned rule, without hurting comfort?

APPROACH

What we built

Simulator

A 1R1C office zone driven by a year of weather, occupancy, price and carbon data

Formulation

An MDP whose reward combines carbon, cost and comfort

Controllers

PPO and DQN against tuned rule-based controllers and an exact DP upper bound

Evaluation

5 seeds, held-out months, 95% CIs, a Pareto sweep, state ablations and policy traces

Responsible AI

What the agent actually learned, and who pays for it

SIMULATOR

A single office zone with a 6-hour time constant

$$C \cdot dT/dt = (T_{out} - T_{in})/R + Q_{int} + Q_{sol} - COP \cdot P$$

- R = 0.4 K/kW and C = 15 kWh/K, so τ = 6 h
- 14 kW HVAC whose COP falls on hot afternoons
- Exact discretisation at 15-minute steps
- Validated by 10 unit tests: step response, energy range, oracle dominance

SERIES	SOURCE
Weather	Singapore-like, 25–33 °C, rainy days
Occupancy	Weekdays 08–18, jitter, holidays
Price	USEP-like, SGD/kWh
Carbon intensity	UK-grid-like (Singapore's ~95% gas grid is nearly flat)

Train January–September · test October–December (92 unseen days)

FORMULATION

The decision problem

ELEMENT	DEFINITION
State	20 features: T_{in}, T_{out}, occupancy sensor, price, CI, time, weekday, forecasts at +1, 2, 4, 8 h
Action	HVAC power $\in$ {0, 25, 50, 75, 100}%, shared by PPO, DQN and the DP oracle
Reward	$-[\lambda_C \cdot CI \cdot E + \lambda_\$ \cdot price \cdot E + \lambda_K \cdot discomfort]$, discomfort counted only when occupied, outside 23–26 °C
Discount	$\gamma = 0.99$, a ~25 h horizon: enough to credit 5 am pre-cooling for 2 pm comfort

Carbon and cost both scale with energy, so being carbon-aware means **shifting when you consume**. Price and CI are nearly uncorrelated here ($r = -0.07$): the cheapest hour is not the cleanest.

CONTROLLERS

Every controller gets the same tuning budget rules

CONTROLLER	BEHAVIOUR	TUNED
Fixed setpoint	Holds 24 °C around the clock	—
Setback	Cools during working hours only	setpoint, start time
Carbon heuristic	Cools deeper when CI is low, drifts warmer when high	4 parameters
Carbon-proportional	Setpoint rises with CI	gain, base
PPO · DQN	Clipped surrogate, GAE, return normalisation	PPO: 12 settings × 2 seeds
Oracle (DP)	Backward induction with perfect foresight	upper bound

All tuning on training days only (972 rule settings), scored with the reward PPO maximises. Tested once on 92 unseen days; 5 seeds per learner, paired 95% bootstrap CIs.

MAIN RESULT

Tuning matters most; PPO's extra gain is small

CONTROLLER	KG CO_2/DAY	VIOLATIONS*
Fixed setpoint	18.96	0%
Setback, 24 °C (untuned)	11.10	0%
Best tuned rule (25.9 °C)	8.52	0.7%
DQN	8.52	7.6%
PPO (tuned)	**8.21**	**5.2%**
Oracle (DP)	7.83	3.7%

*occupied 15-minute steps outside 23–26 °C

−23%

carbon from tuning one setpoint

−3.7%

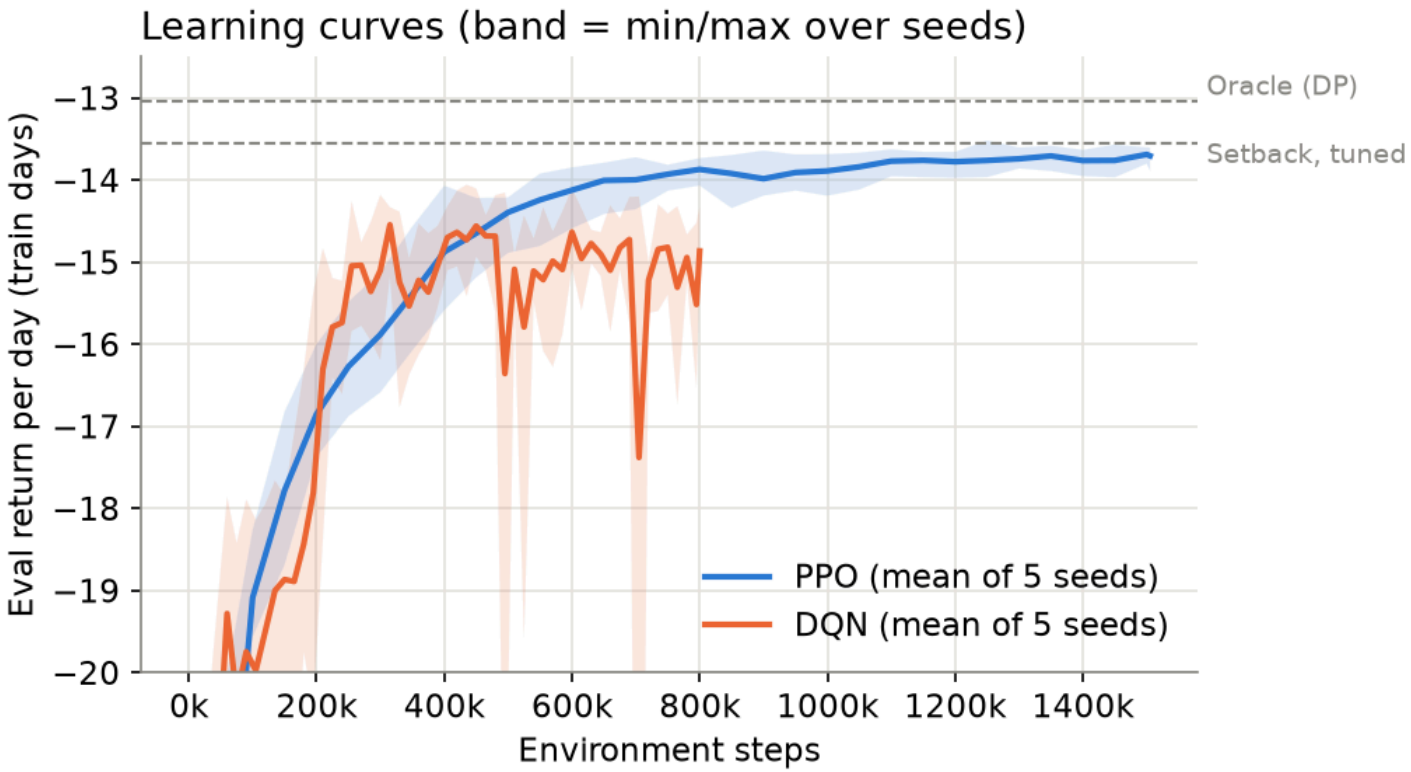
PPO vs best tuned rule, 95% CI [−4.0, −3.2]
reward +1.1% (p = 0.03), cost −4.1%

72%

of that saving is from 3 public holidays, when the rule cools an empty office. Other 89 days: −1.1% carbon, reward −1.6%

LEARNING DYNAMICS

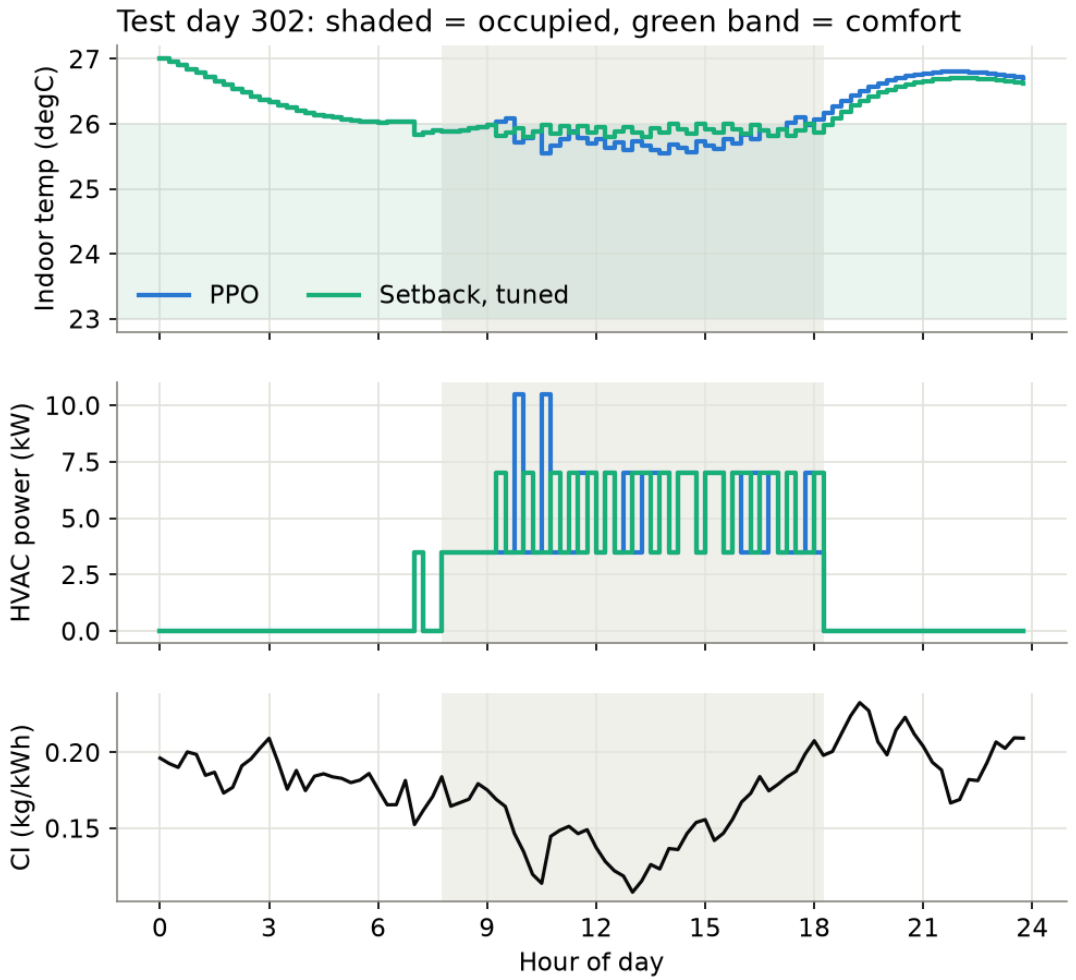
PPO learns stably; DQN is noisy



PPO's 5 seeds settle by 0.75–1.3M steps and land within 1.3% on test carbon. On these 16 training days it ends level with the tuned rule. DQN drops out repeatedly.

POLICY BEHAVIOUR

PPO rides the band edge and cools a little in carbon dips



WHERE THE HEADROOM IS

The bottleneck is occupancy, not the carbon signal

ORACLE VARIANT (TEST)	REWARD	KG CO_2
Full foresight	−13.07	7.83
No carbon foresight	−13.15	8.01
No occupancy foresight	**−14.08**	7.92
Best tuned rule	−13.88	8.52
PPO (tuned)	−13.72	8.21

0.08

reward/day from knowing future carbon intensity

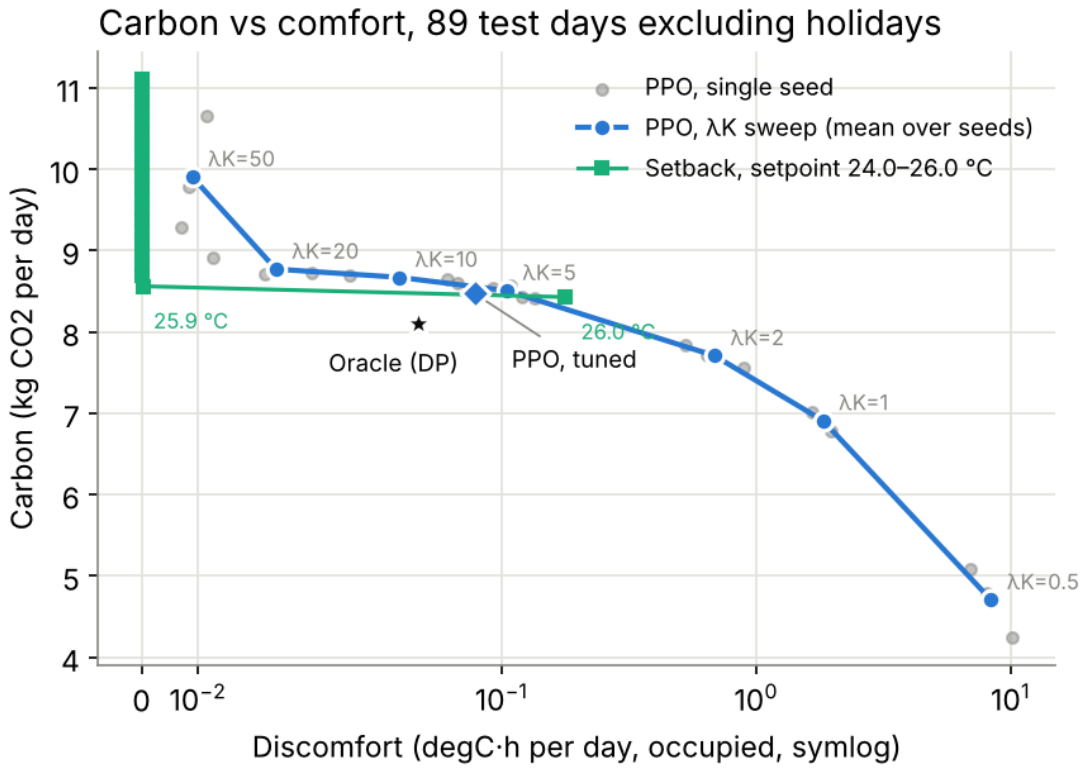
~1.0

reward/day from knowing when people arrive and leave (0.65 on ordinary days)

PPO's own win is occupancy too: it stays off on holidays. Removing CI from its state costs 1.3% carbon (not significant). One room with τ = 6 h is a leaky battery.

TRADE-OFF

At matched comfort, the tuned rule wins



89 ordinary days. At strict comfort every PPO setting is beaten by the rule. PPO only wins at 0.08–0.1 °C·h/day, a band the thermostat skips (25.9 °C → 26.0 °C jumps from 0.7% to 35% violations).

RESPONSIBLE AI

What the numbers do not show on their own

Who sets λ?

The weights decide whose comfort is traded for carbon. The Pareto front makes that trade visible.

Specification gaming

Comfort is charged only when occupied, so a faulty sensor invites exploitation. We score comfort on true occupancy.

Comfort is a distribution

53% of PPO's violations fall between 17:00 and 18:30. Late leavers bear them; the mean hides it.

Honest baselines

An untuned rule inflated PPO's gain to 11%; a holiday calendar shrinks it to 0.9%. Check where a gain comes from.

Transparency

A tuned thermostat gets most of the benefit and anyone can explain it. That is a legitimate choice.

LIMITATIONS

What this study cannot claim yet

Limits

- One zone and a simplified thermal model
- Synthetic profiles, with UK carbon intensity standing in for Singapore's
- A small RL margin; DQN, the λ sweep and the ablations used default PPO settings

Next, for the report

- Occupancy prediction, where the oracle's headroom is
- Robustness: shifted CI, noisy occupancy, tariff regimes
- Constrained PPO: comfort as a constraint, not a weight
- Battery storage: a real lever for shifting carbon

SUMMARY

Takeaways

1. **Tune the baseline first.** One setpoint gives −23% carbon; skipping this inflated our RL gain about threefold.
2. **Tuned PPO wins by 3.7% carbon**, mostly by staying off on holidays; on ordinary days it is ~1%, paid for in comfort.
3. **In one room the carbon signal is worth little.** Occupancy foresight is the real headroom.
4. **An oracle plus fair tuning** is what reveals all of this.

> Thank you. Questions?